DGMS 2019 GEOLOGY • QUESTION 41

Feldspar Petrology: Antiperthite vs. Perthite Exsolution Textures

Topic: Igneous & Metamorphic Petrology | Primary Reference: John D. Winter (Petrology, 2nd ed., Ch. 6)

Question Statement

Q.41. The exsolution texture in which the host mineral is a plagioclase feldspar and the lamellae are of $KAlSi_3O_8$ composition is known as

(a) Perthite
(b) Antiperthite
(c) Myrmekite
(d) Inverted pigeonite

a Perthite	b **Antiperthite (Correct Answer)**
c Myrmekite	d Inverted pigeonite

✓ Correct Answer: (b) Antiperthite [When plagioclase is the host grain and contains exsolution blebs/lamellae of K-feldspar ($KAlSi_3O_8$), the texture is antiperthite; perthite is the reverse].

Textbook Evidence: John D. Winter (Chapter 6)

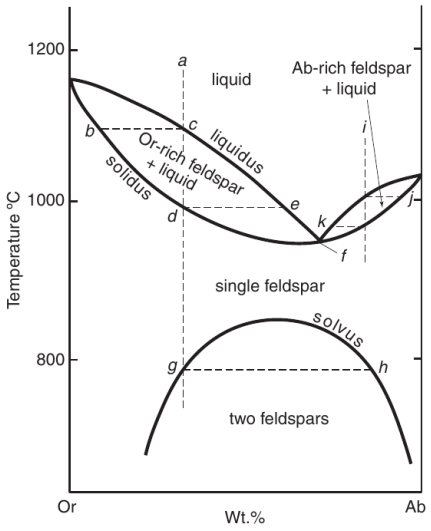


FIGURE 16 *T-X* phase diagram of the system albite–orthoclase at 0.2 GPa H_2O pressure. After Bowen and Tuttle (1950). Copyright © by the University of Chicago.

There is also a **solvus** in this system, one that involves the separation of two *solid* phases from a homogeneous solid solution. The solvus is caused by differences in the size of K^+ ions (ionic radius = 1.59 Å) and Na^+ ions (ionic radius = 1.24 Å). In the pure end-members, this size difference is accommodated by a slightly larger unit cell in orthoclase than in albite. Both structures are stable, but when some of the larger K^+ ions are introfeldspars. When the bulk composition is K rich (as in composition *a* in Figure 16), lamellae of Na-rich feldspar form in a K-rich host, and the texture is called **perthite**. You have probably seen this texture in orthoclase hand specimens. These are the thin, wispy layers seen on a number of cleavages. When the bulk composition is Na rich (as in composition *i* in Figure 16), lamellae of K-rich feldspar form in a Na-rich host, and the texture is called **antiperthite**. Exsolution is less common in plagioclases (probably due to the Ca-Al component and the requirement of coupled Ca-Al for Na-Si exchange). There are three known solvi that occur at low temperatures in the plagioclase system but rarely develop as lamellae and have been left out of Figure 8.

When two feldspars coexist in Figure 16, $\phi = 2$, so $F = 1$. This means that the composition of both feldspars is a function of temperature. As the temperature decreases, the compositions of both feldspars follow the solvus limbs, with the sodic feldspar becoming more sodic and the potassic feldspar becoming more potassic. This is an example of the technique of "geothermometry," by which we can calculate the temperature of equilibration from the composition of analyzed coexisting minerals.

The alkali feldspar system provides another important example of the effects of pressure on mineral systems. The effect of increased H_2O pressure will be greater on the solid–liquid equilibrium than for the solvus because the liquid is both more compressible than either solid and is the only phase that can accept some H_2O. The addition of H_2O would enter the liquid, thereby stabilizing it at the expense of the solid (Le Châtelier's Principle). Increasing the H_2O pressure, therefore, *lowers* the melting point but will have little effect on the solvus.

Figure 17 shows that, as H_2O pressure increases,

Figure 6.16: Subsolidus Solvus Phase Relations in the Alkali Feldspar System (John D. Winter, *Principles of Igneous and Metamorphic Petrology*, 2nd ed., p. 123). As a homogeneous high-temperature feldspar cools below the solvus, difference in ionic radii between K^+ (1.59 Å) and Na^+ (1.24 Å) forces the crystal to unmix into two phases. When the host is K-rich and the exsolved blebs are Na-rich plagioclase, it is termed **perthite**. When the bulk composition is Na-rich and the host is plagioclase with exsolved K-feldspar lamellae, it is termed **antiperthite**.

> *"When the bulk composition is K-rich, lamellae of Na-rich feldspar form in a K-rich host, and the texture is called* **perthite***. When the bulk composition is Na-rich (as in plagioclase), lamellae of K-rich feldspar (KAlSi_3O_8) form in a Na-rich/plagioclase host, and the texture is called* **antiperthite***. Exsolution is less common in plagioclase due to the slower coupled substitution of Ca-Al for Na-Si, making antiperthite especially characteristic of high-temperature granulites and charnockites."*
>
> **— John D. Winter, *Principles of Igneous and Metamorphic Petrology*, Chapter 6, p. 123**

Comprehensive Option Analysis

(b) Antiperthite **[CORRECT]**

Definition & Petrogenesis: Antiperthite is an exsolution intergrowth where:

- **Host Mineral:** Plagioclase feldspar (typically oligoclase or andesine, $\text{NaAlSi}_3\text{O}_8 - \text{CaAl}_2\text{Si}_2\text{O}_8$).
- **Exsolution Lamellae / Blebs:** Potassium feldspar (KAlSi_3O_8, orthoclase or microcline).
- **Geological Occurrence:** Very common in high-grade regional metamorphic rocks of the granulite facies, charnockitic suites, anorthosites, and alkaline plutonic rocks. At high temperatures (750°–900°C), plagioclase can dissolve several percent of the orthoclase molecule; upon slow cooling, the excess KAlSi_3O_8 exsolves as blebs, spindles, or oriented lamellae parallel to (010) or (001).

(a) Perthite **[INCORRECT / REVERSE TEXTURE]**

Perthite is the opposite: the **host is K-feldspar** (orthoclase or microcline) and the **exsolved lamellae are albite/plagioclase** ($\text{NaAlSi}_3\text{O}_8$). It is ubiquitous in granites, syenites, and granitic pegmatites.

(c) Myrmekite **[INCORRECT]**

Myrmekite is a vermicular (worm-like) intergrowth of **quartz enclosed within sodic plagioclase**. It is a replacement/metasomatic texture formed along margins where plagioclase invades and replaces adjacent alkali feldspar, rather than an exsolution texture between two feldspar solid solutions.

(d) Inverted pigeonite **[INCORRECT]**

Inverted pigeonite is a pyroxene texture, not a feldspar texture. High-temperature monoclinic low-Ca clinopyroxene (pigeonite) inverts upon slow subsolidus cooling to orthorhombic orthopyroxene (hypersthene/enstatite) with thick exsolution lamellae of augite (clinopyroxene) oriented along $\{001\}$.

Comparison: Microscopic Exsolution & Replacement Textures

Texture Name	Host Mineral (Matrix)	Lamellae / Inclusions	Formation Mechanism	Diagnostic Rock Types
Perthite	K-feldspar (Orthoclase / Microcline)	Albite / Sodic Plagioclase	Solid-state exsolution below alkali feldspar solvus	Granites, Syenites, Pegmatites
Antiperthite	**Plagioclase (Oligoclase / Andesine)**	**K-feldspar (KAlSi_3O_8)**	**Exsolution of dissolved K-feldspar from high-T plagioclase**	**Granulites, Charnockites, Anorthosites**
Mesoperthite	Equal proportions (~50:50)	K-feldspar and Plagioclase intergrown	Crystallization near minimum of solvus at high pressure	High-grade granulites
Myrmekite	Plagioclase	Vermicular (worm-like) Quartz	Deuteric metasomatic replacement of K-feldspar	Granitoids, Gneisses
Inverted Pigeonite	Orthopyroxene	Augite lamellae along $\{001\}$	Polymorphic inversion of monoclinic pigeonite + exsolution	Layered mafic intrusions (Bushveld, Skaergaard)

Key Takeaway & Exam Strategy

Memory Rule for Feldspar Exsolution:

- **Perthite:** K-feldspar host + Albite lamellae (Think: "P-erthite starts with Potash host").
- **Antiperthite:** Plagioclase host + Potash lamellae (The "Anti" or opposite configuration).
- Host is plagioclase $\rightarrow$ **Antiperthite**!